

Assessment Report

Krishna Kumar

What I Built-

I built NutraReview AI , an AI-powered supplement formulation review system using **Python**, **Streamlit**, and the **Anthropic Claude API**.

The Problem It Solves

Reviewing supplement formulations traditionally requires a team of regulatory experts, toxicologists, and formulation scientists. I built a system that automates this using AI, anyone can paste an ingredient list and instantly get a professional-grade safety and compliance review.

Tech Stack — What and Why

Technology	Why I chose it
Python	Core language, easy to work with APIs and JSON
Streamlit	Fastest way to build a data/AI web app without frontend code
Anthropic Claude API	LLM reasoning powerful enough to understand nutraceutical science
anthropic SDK	Official Python library, supports streaming responses

How the System Works - Step by Step

Step 1 — User inputs

- Product name, category, ingredient list with dosages, and optional marketing claims
- Three sample products are preloaded for quick testing

Step 2 — Prompt Engineering

- The input is structured into a detailed prompt that instructs Claude to act as a nutraceutical scientist

- The prompt explicitly asks for JSON output with a fixed schema — this ensures consistent, parseable results every time

Step 3 — Claude API call with Streaming

- The app calls `claude-sonnet-4-20250514` using the streaming API
- Streaming means the response arrives chunk by chunk, the UI shows a live character count so users know it's working.
- This prevents timeout issues on long analyses

Step 4 — JSON parsing

- The raw response is cleaned (removes any accidental markdown backticks) and parsed with `json.loads()`
- This converts the AI's text output into a Python dictionary the app can work with

Step 5 — Results are displayed

- Overall grade A–F with a score out of 100
- Summary card, strengths and concerns
- Four tabs: Extracted Ingredients, Risk Flags, Formulation Observations, Claims Analysis

AI/ML Components- Most Important

"The requirement was at least one meaningful AI/ML component. I implemented three:"

1. LLM-Based Reasoning

- Claude analyzes each ingredient against known safety thresholds and toxicological data
- It assigns severity levels (low, medium, high) and generates recommendations
- This replaces what would normally require a trained classification model with labeled data

2. Semantic Ingredient Search

- Users type a benefit like *"sleep support"* or *"anti-inflammatory"*

- The LLM understands intent and returns relevant ingredients with mechanisms and evidence levels
- Traditional keyword search would fail here , "sleep" wouldn't find "Ashwagandha" for example

3. Structured Data Extraction

- Free-text like *"Melatonin 20mg, L-Theanine 400mg"* is parsed into structured JSON objects
- Name, dosage, unit, and function are extracted automatically

Deployment

"I deployed it on Streamlit Cloud, which is free and connects directly to GitHub."

Workflow:

1. Code lives on GitHub (public repo)
2. Streamlit Cloud watches the repo
3. Any commit to main auto-deploys within 2 minutes
4. API key- API Keys are paid that's why i did not used the API Key.

You can use your API Key, i am sure it will work properly

Challenges and How I Solved Them

Challenge 1 -Getting consistent JSON from the LLM

- LLMs sometimes add markdown formatting or extra text
- Solution: explicitly instructed the prompt to return *"ONLY valid JSON, no markdown"* and added a cleanup step before parsing

Challenge 2 -Streaming responses in Streamlit

- Streamlit reruns on every interaction which can break streaming
- Solution: used `st.empty()` as a placeholder that updates in place, then clears when done

Why LLM-based reasoning counts as a valid AI/ML component:

- The LLM reasons over ingredient dosages and compares them against its internalized knowledge of safety thresholds — this is not a simple rule-based system
- It understands nutraceutical science, regulatory language, and toxicology without any training data or fine-tuning from my side
- The semantic search understands *intent* ("sleep support" → Ashwagandha, Melatonin, L-Theanine) which is a form of reasoning, not keyword matching

Key Features Delivered

- Extracts structured ingredient data from plain text
- Flags risky/unusual dosages with severity levels
- Generates formulation observations (positive, neutral, concern)
- Detects non-compliant marketing claims (FDA/FTC guidelines)
- Produces an AI-written summary with A–F grade
- Semantic ingredient search by intent

Thank You